

Unit - 3

Electrical conduction in Metals and Semiconductors

➤ Electrical conduction in metals

Metals are good electrical and thermal conductors. The outermost electrons in an atom are loosely bound and are readily shared with other atoms to form metallic bond. Some of these valence electrons are free to move within the physical dimensions of the metal. These free electrons are called conduction electrons because they are responsible for electrical and thermal conduction in metals.

Review of Classical free electron theory (Drude-Lorentz theory)

Assumption of classical free electron theory

1. All metals contain a large number of free electrons which move freely through the entire volume of the metal.
2. The free electrons are treated as equivalent to molecules of an ideal gas and thus obey the laws of kinetic theory of gases. The kinetic energy associated with each electron at a temperature T is given by

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}kT$$

where k is the Boltzmann's constant and v_{th} is the thermal velocity of free electrons.

3. The electric potential due to positive ionic cores is considered to be constant inside the metal.
4. The Coulomb repulsion between the free electrons is considered to be negligible.
5. Electric current flows in a metal in the presence of an external electric field due to drift velocity of the free electrons in a direction opposite to the applied field.

Definitions of some terminology

- **Mean collision time (τ):** The average time taken between two successive collisions of an electron with the lattice points (ionic cores) is called mean collision time. It is given by $\tau = \lambda/v_{th}$; where λ is the mean free path.
- **Mean free path (λ):** The average distance travelled by the electrons between two successive collisions is known as mean free path. $\lambda = \tau v_{th}$
- **Relaxation Time (τ):** The relaxation time is defined as the time during which drift velocity reduces to 1/e times its maximum value after the electric field is switched off.

- **Drift velocity (v_d):** The constant average velocity of free electrons in the steady state in an applied electric field is called the drift velocity v_d .

$$\therefore \mathbf{v_d} = -\frac{eE}{m}\boldsymbol{\tau}$$

-ve sign indicates that drift velocity is opposite to direction of applied electric field.

- **Mobility of electrons (μ):** The ease with which electrons get drifted in the presence of applied electric field is called “Mobility”, denoted by μ . $\mu = \frac{v_d}{E}$;

$$\mu = \frac{v_d}{E} = \frac{eE\tau}{mE} = \left(\frac{e\tau}{m}\right).$$

- **Expression for electrical conductivity (σ):** $\sigma = \frac{ne^2\tau}{m} = ne\left(\frac{e\tau}{m}\right) = ne\mu$

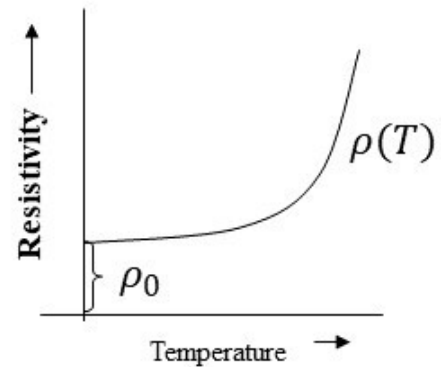
Temperature dependence of resistivity of a conductor or Mathieson’s rule

Experimental temperature dependence of resistivity

Experiments show that resistivity of metals in general vary linearly with temperature in the normal operating temperatures. At these temperatures the resistivity $\rho(T)$ results from the scattering of the free electrons moving with high thermal velocities in random directions.

However, at very low temperatures the resistivity ρ_0 is found to saturate to a non-zero finite value, which is dependent on the impurity concentration in the metal.

Hence, two independent types of resistivity’s determine the total resistivity of the metal due to



1. Scattering of electrons with the vibrating lattice ions - $\rho(T)$

At temperature above zero K, the energy acquired by the valance electrons is sufficient to make them free from the lattice ions. These electrons free to move throughout the metallic crystal leaving behind the positively charged atoms or positive ions. The ions are oscillating about their mean positions due to thermal agitation and are called as lattice vibrations. The

quantized lattice vibrations are called as PHONONS. When free electrons interact with Phonons, they scatter from their paths and accelerate in random direction. This sort of opposition to the flow of electrons gives rise to the resistance. When temperature decreases, intensity of lattice vibrations also decreases, as a result resistivity of conductor also decreases.

2. Scattering of electrons by the presence impurities in the metal - ρ_o

When temperature approaches to zero K, the resistivity of typical conductor reaches to minimum value ρ_o , but it never becomes zero. Here ρ_o is called residual resistivity. This arises due to defects in the conductor and also due to the presence of impurities. Scattering of electrons from the sites of impurities present in the metals such as dislocation joints, grain Boundaries, impurity atom, etc gives rise to resistivity called residual resistivity. It exists even at absolute zero temperature. It is temperature independent.

Thus, the total resistivity of a metal is given by $\rho = \rho(T) + \rho_o$

The above equation is called as **Matthiessen's Rule**, which states that the net resistivity of conductor is equal to the sum of the resistivity due to the phonon scattering which is temperature dependent and resistivity due to the presence of impurity which is temperature independent.

Limitations of classical free electron theory

1. Temperature dependence of electrical conductivity

According to classical free electron theory, $\rho \propto \sqrt{T}$. But, experimentally it is observed that $\rho \propto T$. From this, it is clear that the prediction of classical free electron theory is not in agreement with the experimental observation.

2. Dependence of electrical conductivity on electron concentration (n)

From classical free electron theory, the conductivity $\sigma \propto n$, where 'n' is the electron concentration. But monovalent copper and silver have higher conductivity than the divalent zinc and cadmium, even though the electron concentration in copper and silver is less than that of zinc and cadmium. Hence, $\sigma \propto n$ does not always hold good.

3. The large variation in mean free path also could not be explained by classical theory.

It has been found that mean free path is about one order (ten times) less compared to that of experimental value.

Quantum free electron theory (Sommerfeld theory)

Assumptions of quantum free electron theory

1. The energy values of the free electrons are quantised.
2. The filling up of free electrons in the discrete energy levels happens in accordance with Pauli's exclusion principle.
3. The distribution of electrons in various energy states obeys Fermi-Dirac quantum statistics.
4. The electrons travel in a constant potential inside a metal but stay confined within the boundaries.
5. The attraction between the electrons and the lattice ions and the repulsion between the electrons themselves are neglected.

❖ Comparison

	Classical Free electron theory	Quantum Free electron theory
1	Energy of the conduction electron is continuous	Energy is quantised, $E = \frac{n^2 h^2}{8ma^2}$
2	There is no restriction on the number of electrons possessing a given energy.	Pauli's exclusion principle is applicable. Hence only two electrons with opposite spin states can have the same energy.
3	The energy distribution among the free electrons obeys Maxwell-Boltzmann statistics.	The distribution of electrons in various energy levels obeys the Fermi-Dirac statistics.
4	The electric potential due to ionic cores inside the metal is considered to be a constant.	The electric potential due to ionic cores inside the metal is considered to be a constant.
5	The mutual repulsion between the electrons and the force of attraction between the electrons and ions are neglected.	The mutual repulsion between the electrons and the force of attraction between the electrons and ions are neglected.

- **Fermi energy:** The energy of the highest occupied level at absolute zero temperature is called the Fermi energy and the energy level is known as Fermi level denoted by E_F
- **Fermi factor or Fermi-Dirac distribution function $\{f(E)\}$:** It gives the probability of occupancy of a given energy state by the electrons at thermal equilibrium.

$$f(E) = \frac{1}{e^{\frac{(E-E_F)}{kT}} + 1} \quad \text{where } k \text{ is the Boltzmann constant.}$$

Variation of Fermi factor with temperature

1. Probability of occupancy for $E < E_F$ at $T=0K$

In this case, $(E-E_F)$ is -ve.

$$f(E) = \frac{1}{1+e^{\frac{E-E_F}{kT}}} = \frac{1}{1+e^{-\infty}} = \frac{1}{1+0} = 1$$

i.e. $f(E)=1$ for $E < E_F$, This means all the energy levels below the Fermi level are occupied.

2. Probability of occupancy for $E > E_F$ at $T=0K$

In this case, $(E-E_F)$ is +ve.

$$f(E) = \frac{1}{1+e^{\frac{E-E_F}{kT}}} = \frac{1}{1+e^{\infty}} = \frac{1}{1+\infty} = \frac{1}{\infty} = 0$$

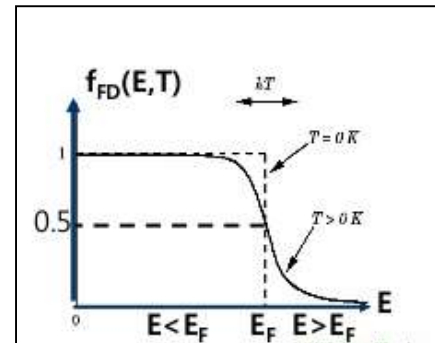
i.e. $f(E)=0$ for $E > E_F$. This means, all the energy levels above Fermi level are empty.

3. Probability of occupancy at ordinary temperature; $E=E_F$ at $T>0K$

$$f(E) = \frac{1}{1+e^{\frac{E-E_F}{kT}}} = \frac{1}{1+e^0} = \frac{1}{1+1} = \frac{1}{2} = 0.5 \quad \text{i.e.}$$

probability of occupation is 50%.

At ordinary temperatures, though the value of probability remains one for $E \ll E_F$, it starts decreasing from one as the value of the E becomes closer to E_F and becomes $\frac{1}{2}$ at $E=E_F$. Further for $E > E_F$, the probability value falls off to zero rapidly.



Density of energy states $\{g(E)\}$

It is defined as the number of available energy states per unit volume per unit energy range centered at a given energy E . It is a continuous function and the product of $g(E)dE$ gives the number of states in the energy interval dE at E .

Expression for Density of states:

We know the allowed energy for a particle confined in one dimensional box can be written as

$$E = \frac{n^2 h^2}{8ma^2} \text{-----(1) where } n=1,2,3,\dots;$$

‘a’ is length over which the particle is free to move in one dimension.

Considering the free electrons in a metal as particles in 3-dimensional potential well, the energy of the free electrons can be written as

$$E = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2) \text{(2)}$$

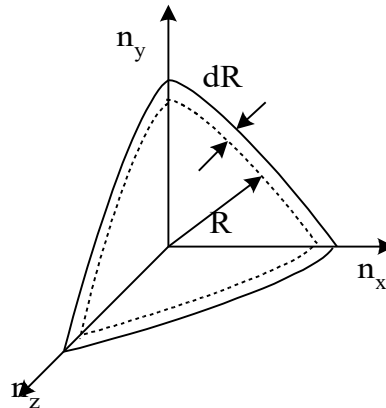
where n_x, n_y, n_z are +ve integers greater than zero.; m --- mass of the electron.

$$\text{Let } E_0 = \frac{h^2}{8ma^2}; \text{ --- (3) and } R^2 = n_x^2 + n_y^2 + n_z^2$$

Then eqn (2) becomes, $E = E_0 R^2$ ----- (4)

The allowed energy states are governed by the values of n_x, n_y and n_z . If we take a coordinate system with n_x, n_y and n_z along three mutually perpendicular axes, then each set of values consisting of three integers can be represented as a point in the system. For a very large number of such sets of values we get a spherical distribution of points in the plot.

As per quantum conditions the values of n_x, n_y and n_z are restricted to be positive. There is only one octant of the sphere in which each point corresponds to only positive values for n_x, n_y and n_z . If R is the radius of the sphere whose octant encloses all the points upto an energy E, then we can write, number of allowed energy values upto an energy E= number of points in the octant of the sphere of radius R.



If we consider a small energy range between E and E+dE, then the number of allowed energy values in the energy range E and E+dE

= number of points in the space between the two octants of radii R and R+dR.

= (volume of space between the octants of radii R and (R+dR) x (Number of points per unit volume)

The number of energy levels having energy values lying between E and E+dE (i.e, in the energy interval dE) is given by volume of the space between two octant shells of radii R and R+dR.

$$\begin{aligned} \text{i.e } g(E)dE &= \frac{1}{8} \left[\frac{4}{3} \pi (R + dR)^3 - \frac{4}{3} \pi R^3 \right] \\ &= \frac{1}{8} \times \frac{4}{3} \pi [(R + dR)^3 - R^3] \\ &= \frac{\pi}{6} [R^3 + (dR)^3 + 3R^2 dR + 3R(dR)^2 - R^3] \end{aligned}$$

Neglecting higher order of dR we get, $g(E)dE = \frac{1}{6} \pi 3R^2 dR = \frac{1}{2} \pi R^2 dR$

According to Pauli's exclusion principle, each energy value is applicable to two energy states, one for an electron with spin up and the other for an electron with spin down.

Hence, number of allowed energy states in the energy range E and E+dE is given by,

$$g(E)dE = 2 \times \frac{1}{2} \pi R^2 dR = \pi R^2 dR \text{-----(5)}$$

Now, differentiating eqn.(4) we have $dE = 2E_0 R dR$ or $R dR = \frac{dE}{2E_0}$ -----(6)

and eqn(4) can be rewritten as $R = \left(\frac{E}{E_0}\right)^{1/2}$ -----(7)

Substituting (6) and (7) in (5) $g(E)dE = \pi \left(\frac{E}{E_0}\right)^{1/2} \frac{dE}{2E_0} = \frac{\pi}{2E_0^{3/2}} E^{1/2} dE$

Substitute, $E_0 = \frac{h^2}{8ma^2}$; we will get $g(E)dE = \frac{\pi (8ma^2)^{3/2}}{2 (h^2)^{3/2}} E^{1/2} dE$

On simplifying we get $g(E)dE = \frac{8\sqrt{2}\pi a^3 m^{3/2}}{h^3} E^{1/2} dE$ -----(8)

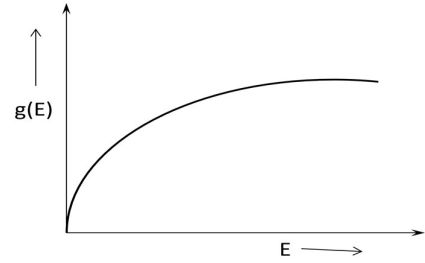
a^3 represents the volume of solid. If we consider the solid to be of unit volume, i.e, $a^3=1$ then

$$g(E)dE = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} E^{1/2} dE \text{-----(9)}$$

This equation represents the density of states.

$$\text{Thus, } g(E) \propto E^{1/2}$$

A plot of $g(E)$ Vs E is as shown. The shape of the curve is a parabola



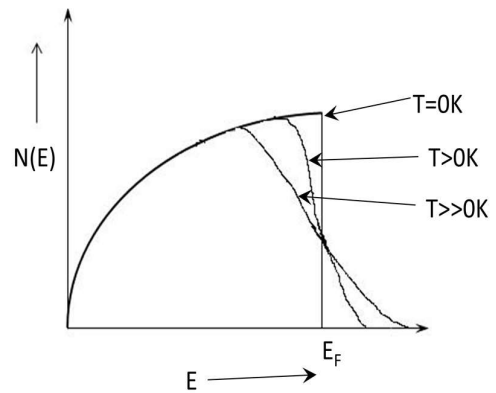
Fermi Dirac distribution

If $N(E)dE$ is the number of those electrons which possess energy only in the range E and $E+dE$, then the number of electrons occupy the energy levels in an energy range dE is,

$$N(E)dE = g(E)dE \times f(E).$$

where $f(E)$ is the fermi factor.

A plot of $N(E)dE$ Vs E , represents the actual distribution of electrons among the available states. The distribution is known as Fermi-Dirac distribution.



Expression for Fermi energy (E_F) at $T=0K$

Let E_{F0} be the maximum energy the electron can possess at $T=0K$, and n be the number of free electrons/unit volume. 'n' can be obtained by integrating $N(E)dE$ in the limit $E=0$ to E_{F0} .

$$\text{i.e. } n = \int_{E=0}^{E_{F0}} N(E)dE = \int_0^{E_{F0}} g(E)f(E)dE,$$

where $g(E)dE = \frac{8\sqrt{2}\pi m^{3/2}}{h^3} E^{1/2} dE$ and at $T=0K$,

For $E < E_F$, $f(E)=1$

$$\begin{aligned} \text{Hence } n &= \frac{8\sqrt{2}\pi m^{3/2}}{h^3} \int_0^{E_{F0}} E^{1/2} dE \\ &= \frac{8\sqrt{2}\pi m^{3/2}}{h^3} \left[\frac{2}{3} E^{3/2} \right]_0^{E_{F0}} \\ &= \frac{8\sqrt{2}\pi m^{3/2}}{h^3} \frac{2}{3} E_{F0}^{3/2} \\ n &= \frac{\pi}{3} \frac{8 \cdot 2^{3/2} m^{3/2}}{h^3} E_{F0}^{3/2} \end{aligned}$$

$$\therefore E_{F0}^{3/2} = \frac{h^3}{(8m)^{3/2}} \frac{3n}{\pi}$$

$$\text{(or)} \quad E_{F0} = \frac{h^2}{8m} \left(\frac{3n}{\pi} \right)^{2/3}$$

This can be written as $E_{F0} = Bn^{2/3}$ where $B = \frac{h^2}{8m} \left(\frac{3}{\pi} \right)^{2/3} = 5.85 \times 10^{-38} \text{ Jm}^2$.

- **Fermi temperature (T_F)** is the temperature at which the average thermal energy of electron in a solid becomes equal to the Fermi energy at 0K.

$$\text{i.e., } kT_F = E_F \quad \text{or} \quad T_F = \frac{E_F}{k} \quad \text{where } k \text{ is the Boltzmann constant.}$$

Fermi temperature is a theoretical concept because even at ordinary temperature it is not possible for electrons to receive thermal energy in a magnitude of E_F .

- **Fermi velocity (v_F)** is the velocity of the electrons which occupy fermi level.

$$\text{Hence, } E_F = \frac{1}{2} m v_F^2 \quad \text{or} \quad v_F = \sqrt{\frac{2E_F}{m}}$$

According to quantum free electron theory, conductivity, $\sigma = \frac{ne^2\tau}{m} = \frac{ne^2\lambda}{m^*v_F}$; where m^* is the effective mass of the electron.

Quantum free electron theory could successfully explain the limitations of classical free electron theory.

Electrical conduction in Semiconductors

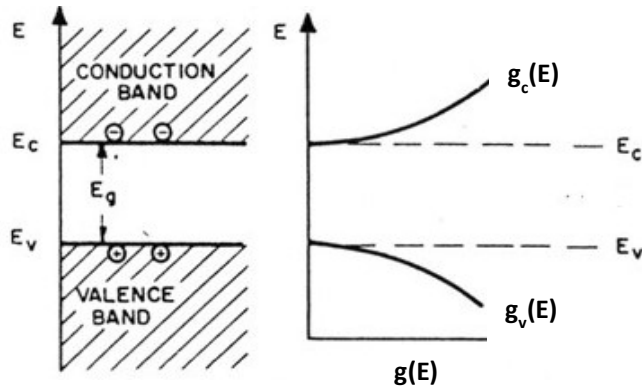
According to the band theory of solids, if the energy gap is quite small as in case of semiconductors, then there is some statistical probability that the electrons will be excited from the valance band to conduction band and will be available for conduction. The energy gap (band gap) of semiconductors is about 1eV. The excitation of an electron into the conduction band results in not only an electron in the conduction band but also a hole in the valence band., which also carries charge. Thus, in semiconductors, both electrons in the conduction band and holes in the valance band are mobile charge carriers which contribute for electrical conduction.

The **density of allowed energy states** for electrons in conduction band is given by

$$g_c(E) dE = \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} (E - E_c)^{1/2} dE$$

where, m_e^* is the effective mass of electron and E_c is bottom most energy level of conduction band.

The figure below shows density of energy states in the conduction band and the density of energy states in the valence band as a function of energy.



➤ **Expression for Electron Concentration in conduction band at thermal equilibrium.**

The density of allowed energy states for electrons in conduction band is given by,

$$g_c(E) dE = \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} (E - E_c)^{1/2} dE$$

The distribution of electrons in the conduction band (CB) is given by the density of allowed quantum states times the probability that a state is occupied by an electron.

$$\text{i.e.} \quad n(E) = g_c(E) dE \times f(E) \quad \text{-----(1)}$$

where $f(E)$ is the Fermi-Dirac probability function.

The **total electron concentration** at thermal equilibrium in the conduction band is then found by integrating eqn (1) over the entire conduction band energy.

$$\text{i.e.} \quad n = \int_{E_c}^{\infty} g_c(E) dE \times f(E) \quad \text{-----(2)}$$

For electrons in the CB, at room temperature, $(E-E_F) \gg kT$. so Fermi probability function reduces to Boltzmann approximation

$$\text{i.e.} \quad f(E) = \frac{1}{1 + e^{\frac{(E-E_F)}{kT}}} \approx \exp\left(-\frac{(E-E_F)}{kT}\right) \quad \text{-----}(3)$$

Substituting (3) in (2) we get,

$$n = \int_{E_C}^{\infty} \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} (E - E_C)^{1/2} \exp\left[-\frac{(E-E_F)}{kT}\right] dE$$

By adding and subtracting E_c in the exponential term, we can write

$$\begin{aligned} n &= \int_{E_C}^{\infty} \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} (E - E_C)^{1/2} \exp\left[-\frac{(E - E_C + E_C - E_F)}{kT}\right] dE \\ n &= \int_{E_C}^{\infty} \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} (E - E_C)^{1/2} \exp\left[-\frac{(E_C - E_F)}{kT}\right] \exp\left[-\frac{(E - E_C)}{kT}\right] dE \\ n &= \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} \exp\left[-\frac{(E_C - E_F)}{kT}\right] \int_{E_C}^{\infty} (E - E_C)^{1/2} \exp\left[-\frac{(E - E_C)}{kT}\right] dE \quad \text{.....}(4) \end{aligned}$$

$$\text{Let us take,} \quad \left(\frac{E - E_C}{kT}\right) = x$$

$$\text{Then} \quad dE = (kT). dx$$

Also, as $E \rightarrow E_C$; $x \rightarrow 0$ and as $E \rightarrow \infty$, $x \rightarrow \infty$

Then eqn (4) becomes

$$n = \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} \exp\left[-\frac{(E_C - E_F)}{kT}\right] (kT)^{3/2} \int_0^{\infty} \sqrt{x} \exp[-x] dx$$

$$\text{But} \quad \int_0^{\infty} \sqrt{x} \exp[-x] dx = \frac{\sqrt{\pi}}{2} \quad (\text{Gamma function})$$

$$\begin{aligned} \therefore n &= \frac{8\sqrt{2}\pi(m_e^*)^{3/2}}{h^3} \exp\left[-\frac{(E_C - E_F)}{kT}\right] (kT)^{3/2} \frac{\sqrt{\pi}}{2} \\ &= \frac{4\sqrt{2}(\pi kT m_n^*)^{3/2}}{h^3} \exp\left[-\frac{(E_C - E_F)}{kT}\right] \end{aligned}$$

$$n = \frac{2(2\pi m_e^* kT)^{3/2}}{h^3} \exp\left[-\frac{(E_C - E_F)}{kT}\right]$$

$$(\text{or}) \quad n = N_C \exp\left[-\frac{(E_C - E_F)}{kT}\right]$$

This is **electron concentration at thermal equilibrium** in the conduction band.

where $N_C = \frac{2(2\pi m_e^* kT)^{3/2}}{h^3}$ and it is called the **effective density of states function for electrons** in the conduction band.

Note: Similarly it can be shown that, at thermal equilibrium **hole concentration** in the valance band is given by

$$p = \frac{2(2\pi m_h^* kT)^{3/2}}{h^3} \exp \left[-\frac{(E_F - E_v)}{kT} \right]$$

$$\text{(or)} \quad p = N_V \exp \left[-\frac{(E_F - E_v)}{kT} \right]$$

where $N_V = \frac{2(2\pi m_h^* kT)^{3/2}}{h^3}$ and it is called the **effective density of states function for holes** in the valance band.

➤ **Law of mass action in semiconductors**

According to law of mass action, “In a semiconductor under thermal equilibrium, the product of electron and hole concentration is a constant and is equal to the square of intrinsic carrier concentration”.

$$\text{i.e.} \quad np = n_i^2$$

The law of mass action holds for both intrinsic and extrinsic semiconductors.

In an intrinsic semiconductor, the electron concentration in conduction band is identical to hole concentration in valance band at any given temperature.

$$n = p = n_i \quad \text{where } n_i \text{ is intrinsic carrier concentration.}$$

According to the law of mass action in semiconductors, if ‘n’ is increased by doping, then ‘p’ decreases such that the product ‘np’ remain a constant irrespective of doping concentration.

The electron and hole concentration in an intrinsic semiconductor is given by

$$n = N_C \exp \left[-\frac{(E_c - E_F)}{kT} \right] \text{-----(1) and}$$

$$p = N_v \exp \left[-\frac{(E_F - E_v)}{kT} \right] \text{-----(2).}$$

For an intrinsic semiconductor concentration of electrons in the conduction band is equal to concentration of holes in the valence band. i.e, $n = p = n_i$, intrinsic carrier concentration.

$$\begin{aligned}\therefore np &= n_i^2 = N_c N_v \exp\left[-\frac{(E_c - E_F)}{kT}\right] \exp\left[-\frac{(E_F - E_v)}{kT}\right] \\ &= N_c N_v \exp\left[-\frac{(E_c - E_v)}{kT}\right] \\ n_i^2 &= N_c N_v \exp\left[-\frac{(E_g)}{kT}\right] \\ \therefore n_i &= \sqrt{N_c N_v} \exp\left[-\frac{(E_g)}{2kT}\right]\end{aligned}$$

where E_g is the bandgap / energy gap of semiconductor.

Thus for a given semiconductor material at a constant temperature the value of $\sqrt{np} = n_i$ is a constant and independent of Fermi energy. It is called the law of mass action in semiconductors.

➤ **Electrical conductivity of a semiconductor**

Consider a semiconductor of length (ℓ) and area of cross-section (A) subjected to an electric field 'E'. Let n_e and n_h be the no. of electrons and holes with drift velocities V_e and V_h respectively.

We know that, current in a conductor is given by, $I = n e A v_d$

Where 'n' is the concentration of charge carriers and ' v_d ' is the drift velocity.

The total current in semi-conductor will be the sum of current due to electrons as well as holes,

$$\text{i.e.} \quad I = I_e + I_h$$

$$\therefore I = n e A v_e + p e A v_h$$

where, n – concentration of electrons in the conduction band and

P - concentration of holes in the valance band

$$\frac{I}{A} = n e v_e + p e v_h$$

$$\text{But} \quad \frac{I}{A} = J \quad (J - \text{Current density})$$

$$\therefore J = e(n v_e + p v_h)$$

$$\frac{J}{E} = \frac{e(n v_e + p v_h)}{E}$$

$$\frac{J}{E} = e \left(n \frac{v_e}{E} + p \frac{v_h}{E} \right)$$

$$\frac{v_e}{E} = \mu_e \quad \text{and} \quad \frac{v_h}{E} = \mu_h \quad (\mu_e - \text{Mobility of electrons})$$

and μ_h - Mobility of holes)

$$\therefore \frac{J}{E} = e(n \mu_e + p \mu_h)$$

$$\frac{J}{E} = \sigma \quad (\sigma - \text{Electrical conductivity})$$

$$\therefore \sigma = e(n \mu_e + p \mu_h)$$

In case of Intrinsic semiconductors, $n = p = n_i$ (n_i - Intrinsic carrier concentration)

$$\therefore \sigma = n_i e (\mu_e + \mu_h) \quad \text{----- (1)}$$

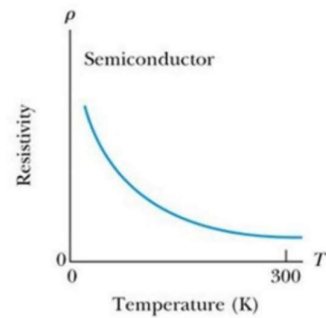
Substituting for n_i , μ_e & μ_h and further simplifying, we get

$$\sigma = C \exp \left[-\frac{E_g}{2kT} \right] \quad \text{----- (2)}$$

where 'C' is a constant for the given semiconductor.

$$\frac{1}{\sigma} = \rho \quad (\rho \text{ is resistivity of semiconductor})$$

$$\therefore \rho = C' \exp \left[\frac{E_g}{2kT} \right] \quad \text{----- (3)}$$



Temperature dependence on resistivity for semiconductors

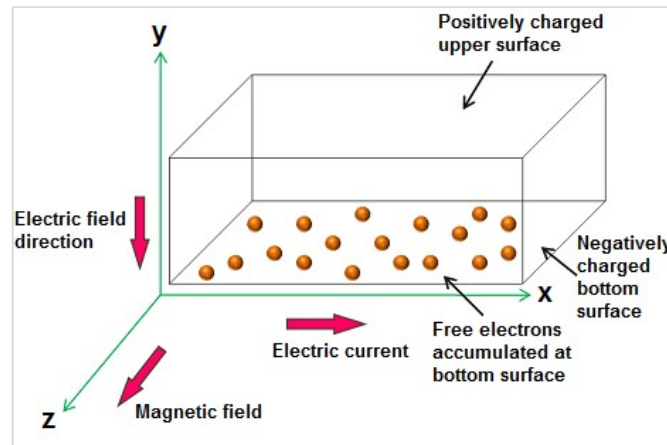
Equation (3) shows that resistivity of a semiconductor decreases exponentially with rise in temperature.

➤ **HALL EFFECT**

When a magnetic field is applied normal to a conductor carrying current, a potential difference V_H is developed in a direction normal to both the magnetic field and current. This is known as **Hall Effect** and generated voltage is called as **Hall voltage**.

Consider a rectangular slab (a metal or semiconductor) carrying current I_x in X-direction. If a uniform magnetic field B_z is applied along Z-direction, then a potential difference is developed along Y-direction (i.e. in a direction perpendicular to both I_x and B_z). This voltage is called Hall voltage (V_H).

Suppose the material is an n-type semiconductor, then current flow is entirely due to majority electrons. Since current is in +ve x-direction, the conduction electrons are drifting with velocity 'v' in the -ve X-direction.



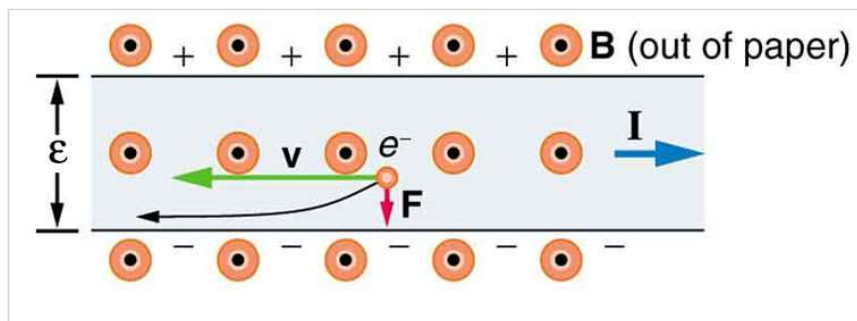
The Lorentz force $\{F_L = q (v \times B)\}$ causes the electron to bend downward as shown in the figure. As a result electrons accumulate on the bottom surface producing a net negative charge there. Simultaneously a net positive charge appears on the upper surface. This creates a downward electric field E_H called **Hall field** and thus Hall voltage V_H is developed between bottom and upper surface.

Expression for Hall voltage (V_H) and Hall coefficient (R_H):

The Lorentz force on moving electrons in the magnetic field acts downwards (as shown in the figure) and is given by,

$$F_L = e (v \times B_z) = e (v B_z \sin\theta) \quad \text{where } v \text{ is the velocity of electrons.}$$

$$F_L = e v B_z \quad \dots\dots\dots(1) \quad (\text{since } v \text{ and } B_z \text{ are perpendicular, } \theta=90^\circ)$$



Now the Hall field (E_H) created by surface charges opposes this Lorentz force. The force due to Hall field acts upwards and is given by,

$$F_H = e E_H \quad \dots\dots\dots(2)$$

In steady state, the net force on electrons is zero.

$$\text{i.e.} \quad F_H = F_L \quad (\text{They are in opposite directions}).$$

$$eE_H = e v B_z$$

$$E_H = v B_z \quad \dots\dots\dots(3)$$

If J_x is the current density in X-direction, then

$$J_x = nev \quad \text{where 'n' is the concentration of charge carriers.}$$

$$(\text{or}) \quad v = \frac{J_x}{ne}$$

Substituting this in eqn. (3), we get

$$E_H = \frac{J_x B_z}{ne}$$

Thus Hall field is proportional to current density and magnetic field.

$$E_H = R_H J_x B_z \quad \dots\dots\dots(4)$$

where $R_H = \frac{1}{ne}$ and is called **Hall coefficient**.

$$\text{From eqn. (4), we can write} \quad R_H = \frac{E_H}{J_x B_z}$$

For n-type semiconductors, electric field is developed in –ve Y- direction. Hence R_H is –ve.

$$R_H = -\left(\frac{1}{ne}\right) = -\left(\frac{E_H}{J_x B_z}\right)$$

In case of p-type semiconductors, R_H is +ve.

$$R_H = \left(\frac{1}{pe}\right)$$

where 'p' is hole concentration.

If 'd' is thickness and 'w' is width of semiconductor, then $A = (w.d)$ is the area of cross section of the slab. If V_H is the Hall voltage across bottom and upper surfaces of slab, then Hall field is

$$E_H = \frac{V_H}{d}$$

From eqn. (4), we get

$$\frac{V_H}{d} = \frac{R_H I_x B_z}{A} = \frac{R_H I_x B_z}{wd}$$

$$V_H = \frac{R_H I_x B_z}{w}$$

$$(or) \quad R_H = \frac{V_H w}{I_x B_z}$$

The quantities V_H , B_z and I_x can be measured experimentally, so the Hall coefficient (R_H) and carrier concentration can be determined.

Applications of Hall Effect:

- **Determination of semiconductor type:** For an n-type semiconductor, the Hall coefficient is -ve whereas for p-type semiconductor it is +ve. Thus the sign of Hall coefficient can be used to determine whether a given semiconductor is n or p-type.
- **Calculation of Carrier concentration:** Hall coefficient ' R_H ' is measured experimentally.

$$R_H = \left(\frac{1}{ne} \right) \quad \therefore n = \left(\frac{1}{e R_H} \right)$$

Thus the concentration of charge carriers can be calculated.

- **Determination of mobility:** If the conduction is due to one type of carriers, eg., electrons, we have $\sigma = n e \mu_n = \frac{\mu_n}{R_H}$

$$\therefore \mu_n = \sigma R_H$$

Thus by knowing σ and measuring R_H , the mobility can be determined.